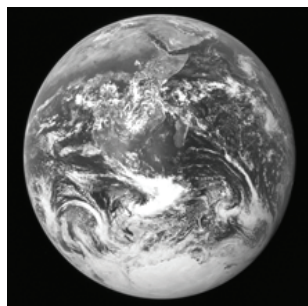


Chapter-3

The Solar System and the Earth

The earth is surrounded by the limitless space. This space includes the stars, galaxy, nebula, comet, planet, minor planet or satellite, aeroliths and other luminaries. The universe has been created with such innumerable luminaries of the stellar sky. The sun is one of the stars of the universe. The solar system or the solar family consists of the sun and its planets, minor planets, asteroids, countless comets and innumerable aeroliths altogether. The sun is the regulator of all planets and minor planets of the solar system. Planets and satellites are attracted by the Sun and their mutual gravitational force, and at certain times the planets revolve around the Sun and the satellites revolve around the planets. The solar system is very small in comparison with the vastness of the universe, and the earth is smaller too. The universe is bigger in size by billion fold in comparison with the earth. In this chapter, we will learn about the concept of the solar system, the planets, the formation of subterranean and time schedule of the universe, the rotation of the earth and its effects, the change of seasons, the surmise of ebb and flow and their influences.



After reading this chapter, we will be able to—

- explain the concept of the universe;
- describe the planets of the solar system;
- explain the causes for habitation of the living beings;
- describe the formation of subterranean;
- show the locations drawing the solar system and the planets as well;
- explain the concept of parallels of equator, latitude and meridians of longitude, prime meridian and international date line;
- assess the role of the imaginary lines in respect of the world's time deciding;
- explain the causes resulting the difference between the time of Bangladesh and that of any other country, and specify the time;
- draw the figures of locations of different lines;
- explain the concept of diurnal rotation and annual motion, and their effects on the earth;
- explain the reason causing waning and waxing of day-night;
- explain the causes resulting in changes of seasons in the world;
- analyse the relation between annual motion and the change of seasons in Bangladesh;
- calculate the time by applying mathematical cognition in a new situation;
- explain the surmise of ebb and flow, their causes and classifications as well;
- explain the effects of ebb and flow on the earth;

Section 3.1: The Solar System

The sun-family comprising the sun and its planets, minor planets, asteroids, comet, aeroliths, is called the solar system. The sun is the centre-point of the solar system. There are 8 planets, more than hundred satellites planets, thousands of asteroids and millions of comets in the solar system.

The Sun

The sun is the regulator of all planets and ‘Satellites’ is sufficient. existing in the solar system. It is a bright star. It has a close tie with the earth. It is greater than the earth by 13 hundred thousand-fold. The earth is away from the sun approximately by 150 million kilometers. It is about 13 hundred 84 thousand kilometers in width. The temperature of its surface is $57,000^{\circ}$ Celsius. A very light temperature from the sun arrives at the earth since the sun stays very far. Such a light temperature and light adequately suffice the need of the living beings on earth. The sources of heat and

light in other planets and minor planets are the sun itself too. The sun has neither any hard nor any liquid substance. The sun comprises of 55% of Hydrogen and 44% of Helium and 1% of other gasses. The black spot that is sometimes seen in the sun is called the Sunspot or Solar infamy. The heat of Sunspot is less than that of its other parts. Under the atomic energy generation process, hydrogen existing in the sun transforms into helium gas, and energy is being generated from this gas in a continuous process. The sun revolves on its own axis once in about 25 days, and it strolls around its own galaxy through the long round way at a gap of about 200 million years. The vibration of life, plants, and animals could not exist in the earth without the light and heat of the sun, and the world had remained in ever darkness.

The Planets: Some luminaries influenced by the gravitation revolve around the sun in the space through certain orbits maintaining a particular duration are called planets. They have no light or heat of their own. The number of planets in the solar system is 8. They respectively stay from the sun in accordance with their distance, namely Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune. Jupiter is the largest among the planets and the Mercury is the smallest planet in the solar system.

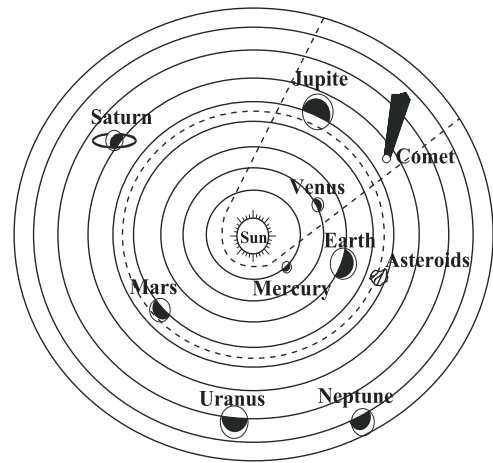


Figure 3.1: The Solar System

Mercury: Mercury is the closest planet to the sun and the smallest planet in the solar system. It is also known as Earth's twin or sister planet. Its diameter is 4,850 kilometers and its weight is equivalent to three-fiftieth of the earth. It orbits the sun once in about 88 earth-days. The average distance of it from the sun is 5.8 crore kilometers. It has no natural satellites. Its temperature is very high since it is the closest planet to the sun. Many holes and mountains with level ground have been noticed on the surface of Mercury. Mercury surface area is 74,800,000 square kilometers.

Venus: Venus is the closest to the Earth and its position is the second from the sun by distance. It is 10.8 crore km away from the sun and only 4.2 crore km from the Earth. We find it in the western sky as the evening star, and at dawn as vesper in the eastern sky. It orbits the sun once in 225 earth days. It has no natural satellites. Like earth, Venus has a substantial atmosphere but has no oxygen in it. The presence of Carbon dioxide gas is about 96%. This planet orbits on its own axis very slowly. Therefore, the sun rises and sets twice a year in its sky. Acid rain showers due to the dense cloud of Carbon dioxide in this planet. It is much drier than the earth, and the pressure of air is ninety times stronger in its atmosphere than that of the earth. Its surface area is 460,230,000 square kilometers and diameter is 12,104 kilometers.

The Earth: The Earth is the third closest planet to the sun. It has an area of 510,100,422 square kilometers. Its diameter

Compare the characteristics of the luminaries in the Solar System in a table

measures 12,752 km in the East-West and 12,709 km in the North-South. The average distance of the earth from the sun is 150 million km. It orbits the sun once in 365 days 5 hours 48 minutes and 47 seconds. This planet has sufficient oxygen and nitrogen as well. The average temperature of the surface of the earth is 13.90°C. It has necessary water on its crust. Of all the planets, the Earth is the only planet which is ideal for living for the creatures and plants. It possesses one natural satellite, the moon, The average distance of the Earth from the Moon is 381,500 km.

Mars: Mars, after the Earth, stands close to the Sun by distance. Its average distance from the sun is 22.8 crore km and 7.8 crore km from the Earth. Its diameter is 6,779 km and its weight is equivalent to one-tenth of the Earth. It has an area of 144,798,500 square kilometers. It takes 687 days to orbit the Sun, and it orbits once on its own axis in 24 hours and 37 minutes. It has two tiny natural satellites (Deimos and Phobos). Living is impossible here. It possesses an

atmosphere containing 3% of nitrogen and 2% of argon gas. The amount of water is very scant here. It is colder than the Earth and its average temperature is below the freezing point. Its surface is peppered with gorges and vast volcanoes. Its red color comes from iron oxide (rust) in its soil.

Asteroids: Numerous small planets are orbiting together in between Mars and Jupiter. There is no other planet in this extent. A luminary with a diameter less than 1.6 km to 805 km is called 'asteroid'. They are called asteroids in a body.

Jupiter: Jupiter is the largest planet of the solar system. It stands fifth from the sun by distance. Its measure is equivalent to about 1300-fold of that of the Earth that is, it has an area of 61,419,000,000 square kilometers. Its diameter is 1,42,800 km. It is about 77.8 crore km away from the sun. Jupiter orbits the sun once in 12 years and on its own axis in 9 hours and 53 minutes. The sun rises and sets in this planet twice in one earth-day. It possesses a dense atmosphere. The up-level temperature of this planet's surface is very low and its internal temperature is very high. It has 97 satellites of which Ganymede, Callisto, Io and Europa are the most important.

Saturn: Saturn is the second largest planet of the solar system. It has an area of 42,700,000,000 square kilometers and diameter of 116,464 km. It is 143 crore km away from the sun. It orbits the sun once in 29 years and 5 months and on its own axis once in 10 hours and 40 minutes. Saturn is larger than the Earth by 900 fold approximately, and it can be watched with open eyes. It possesses an atmosphere containing mixture of hydrogen and helium, methane and ammonia gases. Three bright belts have surrounded Saturn. It has 62 satellites of which Capitas, Tethis, Hua, Titan, Enceladus are to be mentioned.

Uranus: Uranus is the third largest planet. It is 287 crore km away from the sun. It orbits the sun once in 84 years. Its average diameter is about 49,000 km. Its volume is larger than Earth by 64-fold and its weight is heavier than that of the Earth by 15-fold only. Its atmosphere possesses a vast amount of methane gas. It has 27 satellites of which the largest ones are Miranda, Ariel, Oberon, Unbriel and Titania, etc. It is discovered that Uranus has belts like Saturn.

Neptune: Neptune's average diameter is 49,244 km. It is 450 crore km away from the sun. It has an area of 17,618,300,000 square kilometers. This planet is cold since it stays much far away from the sun. It is largely bluish to look at. It orbits the sun once in 165 years. It has 14 satellites -Triton and Nereid are its significant satellites. Neptune is accompanied in its orbit by a number of minor planets termed Neptune Trojans.

The Causes for Habitation of Living Beings on the Planet Earth

All sides of Earth are surrounded by miscellaneous gaseous elements. Such invisible gaseous covering is called the atmosphere of Earth which is implicated in the surface of the Earth being attracted by the gravity, and being rotated along with the Earth. Its density is most on the sea-surface due to air-pressure, and its up-level density is comparatively less. Nitrogen and Oxygen have supremacy in the atmosphere. Oxygen is a must for all kinds of creature. The extent of other elements other than that of Carbon dioxide gas remains almost invariable, but some elements like dust, smoke, aqueous vapour, etc. remain at particular places in different extent. The atmosphere plays a vital role in order to help all creatures live in the Earth. It protects the fauna from the harmful rays of the sun, and its gaseous element namely Carbon dioxide (CO_2), saves plants and oxygen (O_2) saves the fauna as well. Troposphere is the level adjacent to earth-surface and is the lowest level of the atmosphere. This level possesses an average depth of about 13 km. It is the most essential level for human beings, because moistness, clouds, fogs, rains, streams of air, etc. are noticed at this level. Due to the increase of height, the velocity of air gets enhanced at this level, and air moves up and down. Most of all the processes of the climate and weather happen at this level of atmosphere. The atmospheric boundary between the Troposphere and the Stratosphere is called Tropopause. The depth of Tropopause is narrow, and the air remains standstill there, and spacecraft routinely pass through this level without any trouble for not having any prevalence of rainstorms. Ozone gas possesses a level in the atmosphere known as Ozone level. Its depth is approximate 12-16 km. Since it absorbs the ultra-violet ray of the sun, temperature reaches at about 40°C . This level of the atmosphere has made the Earth habitable for the living beings.

Individual work: Make a list showing reasons why other planets are not suitable for habitation except the Earth.

The Earth receives light and heat from the sun. This Earth would remain ever dark without the light of the sun. No life vibration could remain in the Earth, and all creatures, plants and fauna could not live. The human activities on Earth change the formation of the atmosphere and its elements. For example, deforestation, smokes emitted from the mills and industries, and burning of fuels, coals and natural gases enhance the pollution of the atmosphere. It is necessary to keep the atmosphere fresh for the sake of existence of the living beings. The humans and animals need light, air and water in abundance. The average temperature on the surface is 13.90°C . Its surface possesses sufficient water. The heat and light that reaches the Earth from the sun are also endurable by the fauna. These are very essential and suitable elements for the existence of the flora and fauna as well. That is why the humans and animals can live comfortably on Earth.

Interior Structure of the Earth

It is very difficult to collect actual information about interior structure of the earth and its various layers. But Earth's interior rock's density-variation and different layer's characteristics can be known by various seismic wave velocity of interior earth. So the Geologists have divided interior structure of the earth into three layers on the basis of velocity and nature of seismic wave of interior of earth. These layers are- (1) Core, (2) Mantle and (3) Crust.

(1) Core: The radius of round shape earth is 6371 km (approx.). Around the earth's centre, there is a spherical object having 3486 km (approx.) radius. This spherical object is defined as core. Specific gravity of this layer is 10 to 13.6. According to the scientists, core is composed of some solid and heavy materials nickel and iron (Ferrum) portion. It is 10/12 times dense more than water and two times than other portion of the earth. Probably it is not in solid form because of excess heat and pressure. Based on seismic wave, core is divided into two part- external part and internal part. It is assumed that outer part of core is liquid and inner part is solid. The extent of outer portion of the core is near about 2270 km. Inner portion of core is still solid having 1216 km radius from earth centre and it is mainly composed of iron and nickel.

(2) Mantle: The surrounding up to 2885 km. radius from upper level of the core is defined as mantle. This layer contains heavy metals like silicon and magnesium. As its upper layer 1448 km. composed of basalt like materials, it is called basalt region. This layer is also named as Sima because of its composition of silicon and magnesium.

(3) Crust: The upper part of the mantle is called crust. It is composed of various rocks and minerals. Its depth is the highest in the beneath of continental region and the least in oceanic region. There is a debate about its actual depth. Its depth is considered up to 30 to 64 km. according to location. The mentionable elements of the composition of this layer are oxygen, silicon, aluminium, iron, calcium, sodium, potassium, etc. It is mentionable that Silicon and Aluminium are abundant at this layer. That is why this layer is called SiAl layer. The upper part of

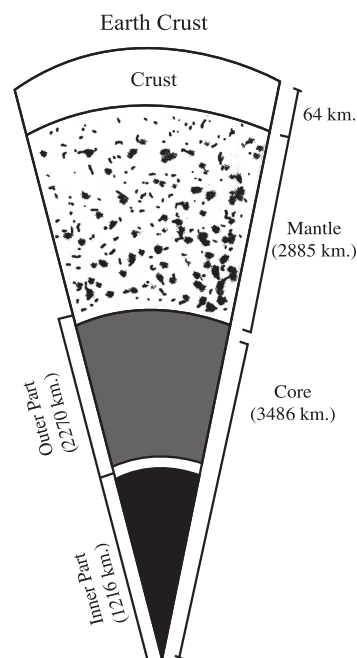


Fig. 3.2: Interior Structure Of the Earth

the crust is called earth-crust and lower part of the crust is called substratum. Earth-crust is hard portion of the earth. Its depth is 3km.(beneath the ocean) to 40 km.(beneath the mountain) but average depth is 17 km.

Section 3.2 : The Method of Calculating Time at Different Regions of the World

A few imaginary lines are drawn in the world's map across the East-West and the North-South in order to decide location of any region in it. These are respectively called Axis and Meridians of Longitude. The location of any region can be known through the Axis and the meridians of longitude. Any regional time can be known from the position of the meridians of longitude. As the locations in the North and South can be known from the equator through the axis, the meridians of longitude is used to know about the locations in the East or West from the prime meridian as well. Though the Earth's surface seems level, it is an approached globe in fact. That is why the distance from the prime meridian can be easily demonstrated in an angular measure as the Earth is almost round.

The Axis, Equator, Latitude, Meridians of Longitude and Prime Meridian

The Axis and Equator: The line imagined at the North-South over the center of the Earth is called Axis. Its northern end point is called the North Pole or the Arctic, and its southern end point is called the South Pole or the Antarctic. The line that is imagined hemming the Earth across the East-West by keeping the same distance from both the poles is called Equator or Equinoctial Line. This line is circular in size for round shape of the Earth. So it is also called Equinoctial Circular. The Equator has divided the Earth equally into two parts across the North-South. A half of the Earth existing at the North to the Equator is called northern hemisphere, and the other half of the Earth at the South to the Equator is called southern hemisphere. An angular distance of any region within northern or southern hemispheres can be decided through the Equator. A unit of latitude measure like the geometrical angle is called Degree.

The Parallel of latitude: An angle of 360° originated at center of the Earth's circular is divided into Degree (°), Minute (') and Second ("). Angular distance of each pole from the Equator is 90° . A parallel line of the Equator which is imagined by dividing this angle into degrees and minutes is called a parallel of latitude. These parallels of latitude are stretched out towards the East-West, reciprocally parallel, each is a full circle, if latitude increases, the circumference also increases, and the maximum latitude is 90° . The well-known parallels of latitude are $23^\circ 5'$ north latitude known as the Tropic of Cancer, $23^\circ 5'$ south latitude known as Tropic of Capricorn, $66^\circ 5'$ north latitude known as The Arctic Circle, and $66^\circ 5'$ south latitude known as The Antarctic Circle. The angular distance of any region

situated either at the North or the South from the Equator is called the latitude of that region. In order to know the location of any region, it is necessary to identify where the region exists and how far it is in the North or South from the Equator, or in the East or West from the prime meridian. All regions situated at one Axis have the unique latitude. A latitude possessing 0° to 30° is called the Lower Latitude, 30° to 60° the Middle latitude and 60° to 90° the upper latitude.

The Meridians of Longitude

The lines that are imagined from the North to the South pole on each division of the equator into Degrees, Minutes and Seconds are called the Meridians of Longitude. It is also named the meridian. The Meridians are semi-circle and not parallel. Each of the meridians of longitude possesses the equal length. The highest meridian becomes 180° . Considering any of the meridians of longitude a particular prime meridian, an angular distance of the other meridians of longitude from this line, can be measured. The Local time is decided through the meridian.

The Prime Meridian

There is an international agreement that the meridian running through Greenwich, London of the UK over the Royal Observatory in Greenwich at a vicinity of London city, which is spread out over the North Pole and the South Pole is considered the official Prime Meridian. Any line of the longitude (a meridian) can serve as the 0°

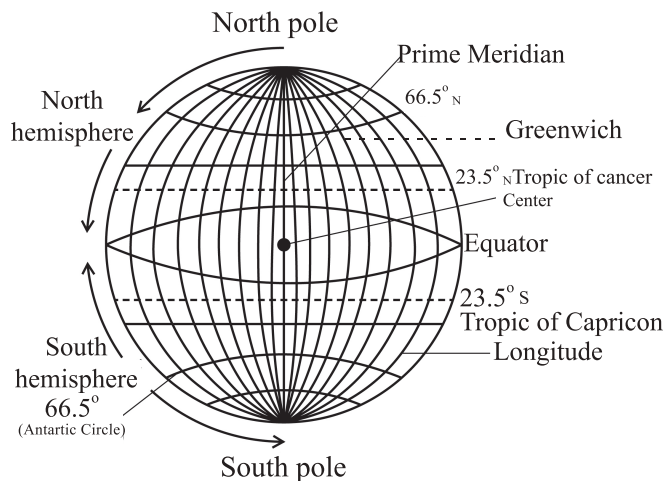


Figure 3.3 : Important Geographical meridians

longitude line. The other meridians can be drawn from the Prime Meridian by the angle originated at the Earth's centre. The meridian existing at the East by 45° from the Greenwich Prime Meridian, Meridian of all region over it is 45° Eastern

Meridian. We can say that an angular distance of any region in the East or West

Individual Work: Note the situations of the important imaginary lines (in 0 degree) in a table.

from the Greenwich Prime Meridian is the Meridian of that region. We also know that Greenwich meridian serves as the Zero(0) degree. Any angle originated with the diameter of the Earth possesses 360 degrees. The Prime Meridian has divided these 360 degrees equally into two divisions after every one degree i.e. by 180° in the East and 180° in the West. Since the Earth possesses a round shape, both the 180° East Meridian and the 180° West Meridian remain within the same meridian. Like the Latitude, the meridian is divided into minutes and seconds as well. Each minute meridian is equal to $1/60$ of a degree. Where there the Equator and the prime meridian intersect each other lengthways, both the latitude and meridian fall in Zero (0) degree. That is the region of the Gulf of Guinea. It is possible to determine the meridian of any region from the Local time and the Greenwich Mean Time.

The Difference in Local Time: The earth is round and it continuously moves around its own axis from the West to the East. So the different regions of the Earth face the sun at different duration. When the prime meridian of any region reaches just the front of the sun, the sun is found exactly overhead at that region, and then, the midday occurs at that region and it is 12 o'clock by watch over there. The other timings of the day are decided in accordance with the midday. We know that one degree of meridian difference results in difference of time by 4 minutes, and the distance of one minute meridian results in difference of time by 4 seconds. When it is 12 o'clock at any place or region, the time of any region existing 5° East to that place or region, will be $12 + (5 \times 4)$ minute or 12 hours 20 minutes. The time of any region existing at 5° West to the same region, will be $12 - (5 \times 4)$ minute or (12H-20M) 11 hours and 40 minutes.

Through Greenwich Mean Time: The Greenwich Meridian serves Zero (0) degree. The exact time of Greenwich can be known by the Chronometer watch. The meridian of a place is calculated by the sextant apparatus, and the midday of that region at that time is determined by observing the highest position of the sun in the sky. The meridian of such region is ascertained from the difference of time between Greenwich Mean Time and the time of that region at a particular duration. If any region is situated to the eastern side of Greenwich, the local time of that region will be more than that of Greenwich, and if it is situated to the western side thereof, its local time will be less than that of Greenwich.

The Local Time and the Standard Time

Local Time: Every day the Earth orbits its own axis from the West to East regularly. Consequently, the sun rises earlier in the regions located at the eastern part of the Earth. In the course of Earth's rotation, when the sun remains overhead or at the highest site, midday occurs at that region and it is 12 o'clock by watch over there. The other timings of the day are determined from such meridian while. It is termed as the local time of that region. The local time is decided by the sextant apparatus too. The measure of angle at the Earth's centre is 360 degrees. The earth orbits such 360° angular distance once in 24 hours or $(24 \times 60) = 1,440$ minutes. So the Earth orbits 1 degree in $(1,440 \div 360) = 4$ minutes i.e. time differs by 4 minutes due to the difference of every one degree meridian.

The Standard Time: If the local time is determined by ascertaining the duration of midday sun over the meridians of longitude, the calculation of time faces confusion in the same country. For that reason, a separate standard time for each country is determined where necessary. In every country, the time that is determined as per the meridian at any region of its center is called the standard time for that country. If the country is larger, it may have several standard times itself. Such as The United States of America have 4 standard times, and Canada has 5. The local time of Greenwich (0° meridian) has been accepted as the standard time for the whole of Earth. The Standard Time of Bangladesh is +6 hours from the Greenwich Mean Time. 90°

East meridian is situated almost above the middle of Bangladesh territory. For this reason, the local time of this meridian is considered the standard time of Bangladesh accordingly.

Individual work: Decide local time and standard time.

The difference of time between Dhaka and Seoul is 2 hours 32 minutes. What is the meridian of Seoul while it is situated at the 90° East to the meridian of Dhaka (Seoul is situated to the East of Dhaka)?

The meridian of Seoul will be more since it is situated in the East of Dhaka.

The difference of time between Dhaka and Seoul is 2 hours 32 minutes = 152 minutes.

The Meridian differs by 1° at a difference of 4 minutes.

So the meridian differs at a difference of 1 minute by $(\frac{1}{4})^\circ$

So the meridian differs at a difference of 152 minutes by $(\frac{1 \times 152}{4})^\circ = 38^\circ$

So the meridian of Seoul stands $90^\circ + 38^\circ = 128^\circ$

Answer: The meridian of Seoul stays at 128° East.

♦ The Meridians of Dhaka and Chennai are respectively 90° East and $80^\circ 15'$ East. What is the local time in Chennai when the midday occurs in the sky of Dhaka?

The difference of meridian between Dhaka and Chennai is $90^\circ - 80^\circ 15' = 9^\circ 45'$. Time will vary for $9^\circ 45'$ meridian, and the time differs by 4 minutes for 1° meridian's difference.

So, the time variation for 9° meridian is $9^\circ \times 4'' = 36$ minutes. Time differs by 4 seconds due to $1'$ meridian's difference.

So the time difference due to $45'$ of the meridian by $45' \times 4'' = 180$ seconds = 3 minutes. So the time difference due to $9^\circ 45'$ meridian by 36 minutes + 3 minutes = 39 minutes.

Chennai is situated at the West of Dhaka (Chennai's meridian is less). Accordingly, its time is less.

So, when the midday is in Dhaka, i.e. 12 o'clock at noon, the local time of Chennai will be 12 hours 39' minutes – 11 hours 21 minutes a.m.

Answer: The local time of Chennai is 11 :21 a.m.

The Antipode

A point that stands opposite to any point existing on the Earth's surface is called the Antipode of that point. The Antipodes exist entirely to the opposite of one another. In order to decide antipode, an imaginary line from any point of the surface through the Earth's center is drawn towards the side contrary to the Earth. The point where the so supposed line reaches at the opposite side of the surface, the very same point is the Antipode to the previous point (see figure). If a latitude of any region is known, its Antipode's latitude is also known as well. The degree of a latitude of any region is the degree of its antipode's latitude. Either of both the regions will be situated at the North and at the South of the Equator. Both the regions will be in both the hemispheres. If one region's latitude stands at 70° North, its Antipode's latitude will be at 70° South as well. If a meridian of any region is added to its antipode's meridian, the summation will be 180° . So, if the meridian of one region is deducted, its antipode's meridian is got. If the meridian of one region stands at the East, its antipode's meridian will be at the West. For example, a meridian of antipode of a region is

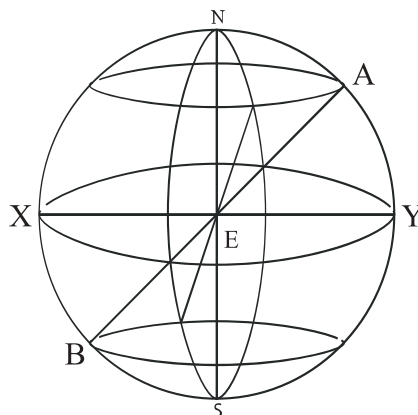


Figure 3.4: The Antipodes
Antipode of place X is Y
Antipode of place A is B

situated at 40° of the East meridian, its will be $180-40^\circ = 140^\circ$ to the West. The time will differ in between both the regions by 12 hours. In the figure 3.3, the antipode of A point is B point (see figure). The antipode of Dhaka is situated over the Pacific Ocean near Chile within the territory of South America.

The International Date Line

While traveling a long way from any particular region to the East or West, days of weeks sometimes get confused along with the local time difference. Problems arise to pass through 180° meridians of longitude to the East or West from any specific venue. In order to solve such problems, a line extending across the North-South over the water surface is imagined by having the recourse to 180° meridians of longitude. This imaginary line is named 'The International Date Line'.

The Importance of the International Date Line: We know that the time differs by 4 minutes for 1° of difference in the meridian. So, the time difference will be 1 hour for 15° of difference in the meridian. If it (the meridian of Greenwich) advances forward from the prime meridian to the East, 12 hours will be more at 180° meridian, and if it gets ahead towards the West, 12 hour time will be less at 180° meridian. So, when it is 10 o'clock on Monday at the prime meridian, the local time will be 10 o'clock night on Monday at 180° East meridian as well. Accordingly, if the meridian of longitude is calculated by the Western side, the local time will be 10 o'clock at night on Sunday, i.e. the previous day at 180° West meridian instead. But the line at 180° East and 180° West is the same. So, it is observed that the local time differs at the same meridian by 24 hours or a whole day. At the same region, where there is Sunday, there is Monday elsewhere, but at the same meridian, 10 o'clock on Sunday

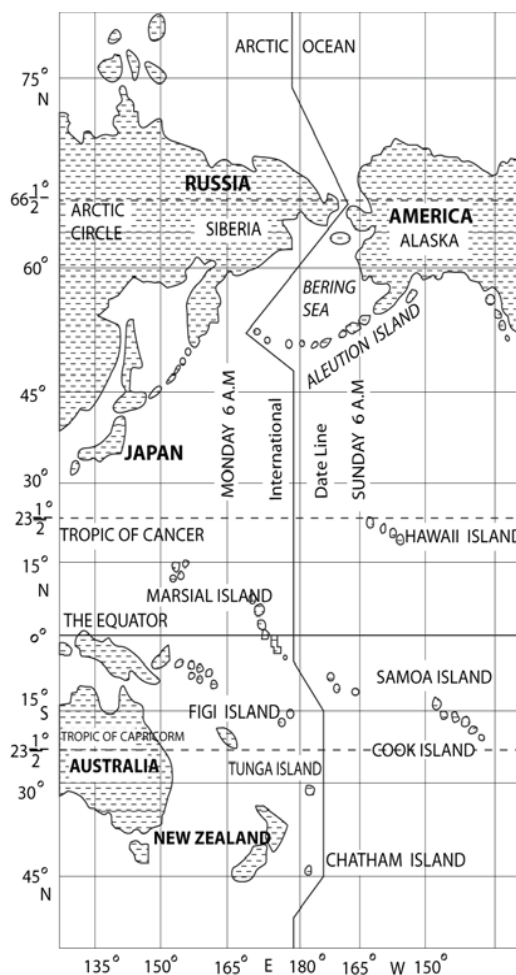


Fig.3.5: International Date Line

night and 10 o'clock on Monday night cannot happen at the same time. In order to solve this problem, a line that has been imagined having recourse to the 180° meridian in the atlas over the water-surface of The Pacific Ocean under the International recognition is the International Line. Since passing through this line results in changes of day and date, that's why it is called International Date Line (see figure). It passes through the northeast region of Siberia, Allucian, Fiji and Chatham islands. In order to solve problems of the local people in calculating 30° dates, this line has been designed over the water-surface in Bering system making it bent towards at 12° East Allusion islands 7° West, and 11° East at Fiji and Chatham islands.

By ship or aircraft towards eastern region, while people routinely pass through this line, they deduct one day from their extended time in order to maintain harmony with the local time over there. Similarly while passing through it towards western region, people calculate their date by adding one day to their curtailed time.

The Role of Imaginary Line to Determine Time

The Earth is almost like a globe. So the imaginary line plays a great role in the world's map in order to determine time within its territory. The round shape Earth orbits its own axis from the West to the East in due course. Consequently, different parts of the Earth's surface face the Sun at different durations. While the meridian of any part stands just in front of the sun at a particular time, the midday occurs at that part and it is 12 o'clock by watch over there. The other timings of the day are determined in accordance with the Noon or Midday. The Earth orbits from the West to East, so the sun rises earlier in the parts existing in the eastern side. If it is 1 o'clock, the time at its 1° East will be 4 past one, and at 1° West 4 minutes to one. In Greenwich (0°) when it is 8 am, it is 10 am somewhere else, and the meridian of such place will be Meridian 30° East. Again, when the time runs behind than that of Greenwich, the said place will be at the West of Greenwich.

Section 3.3: The Motion of Earth

We regularly notice that the sun rises in the East and sets in the West in due course. But have we ever thought on how it happens? It is because the Earth is on motion. The Earth being regulated by the gravity, rotated on its own axis following the sun in its front and moves round the sun through

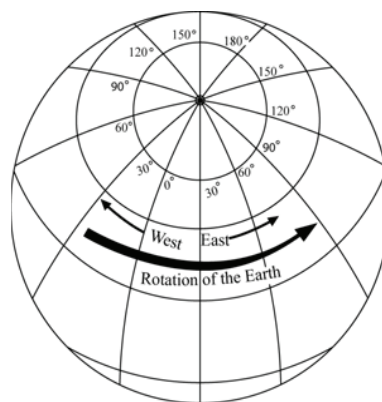


Figure 3.6: Earth's Rotation from West to East

particular orbits. This is the Earth's motion. The Earth's motion is classified into two ways - The Annual Motion and the Diurnal Motion.

The Diurnal Motion: Let us have a look to the figure 3.9. What do we see there? There are a burning candle and a globe. We can see and notice that one side of the globe is illuminated and the other side is dark. The day and night come alternatively in the same manner due to the diurnal motion in the Earth. The Earth rotates on its axis from the West to the East regularly. It rotates in this way almost 24 hours or a day. The time for one complete rotation is 23 hours, 56 minutes and 4.09 seconds. This motion of Earth is called the 'Diurnal Motion'. The time for one complete rotation of Earth is called a Solar Day. The days and nights come due to the Diurnal Motion. The Earth possesses no light itself. It gets illuminated with the sunshine. Since the Earth is round, the sunlight cannot reach the whole of its surface at a time. During the rotation, the part where the rays fall in, day occurs, and the other part remains dark, and vice versa. The time can be calculated on recourse to the diurnal motion. Considering the complete rotation 24 hours, the time is calculated by dividing it in hours, minutes and seconds as well. The ebb and flow takes place in the Earth by attraction of the Moon and the Sun due to the diurnal motion. The diurnal motion affects to a great extent the sea-streams and air flows.

The Experiment: Suppose a burning candle on the table in a dark room as the Sun and the globe as the Earth, we can find that the front part of the globe is lighted and its opposite part remains dark by moving the globe before the burning candle. Day occurs in the lighted part and night in darkness. The circular region in between the lighted and the dark regions of Earth is known as the shade-circle. The part of the Earth that just reaches the lighted part due to the rotation by passing through the shade-circle from the darkness dawn occurs over there. The time when meager light exists sometimes earlier from the daybreak is called the dawn, and the time when meager light exists before sunset is called twilight.

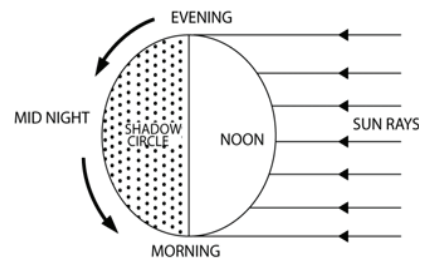


Figure 3.7: Occurrence of Day-night



Figure 3.8 : With the candle and the globe

The Annual Motion: The Earth moves round the sun continuously on its own axis through a certain semi-circular orbit, to a certain direction and in a certain duration. Such rotation of Earth to the East from the West is called the Annual Motion. The Earth moves round the sun in a speed of 30 km per second. The time required for the earth to make one complete rotation around the sun is 1 year. It is called the solar year. The exact time is 365 days, 5 hours, 48 minutes, 45°51 seconds. As the solar year is calculated on 365 days, every fourth English year is calculated with an additional day, i.e. 366 days. In that year, February possesses 29 days instead of 28, and such year is called the 'Leap Year'. The waning and waxing of the day-night and the changes of seasons happen in the Earth in consequence of the annual motion.

The Role of the Annual Motion to the Waning and Waxing of Day-Night and to the Changing of the Seasons

We have noticed that the time of different days and nights in the year have difference, i.e. sometimes days are longer and sometimes nights are longer. Have we ever thought on how it differentiates? It can be replied that such disparity occurs due to the annual motion. An annual motion causes days and nights and to change the seasons.

The Reasons that Make Days and Nights Longer or Shorter

Let us have a look to the following figure. While moving round the sun, the four stays of the earth have been specified in the orbit such as June 21, September 23, December 22 and March 21.

June 21: While moving round the sun, the earth upon its course reaches at such a place on the orbit on June 21, where the North Pole mostly inclines ($23^{\circ}5'$) towards the sun, and the South Pole remains farthest away from the sun. On this day at noon, the sunlight falls over $23^{\circ}5'$ North Latitude lengthways (at angle 90°). The day on this date becomes the longest in the north hemisphere and the night becomes the shortest. The opposite situation exists in the south hemisphere

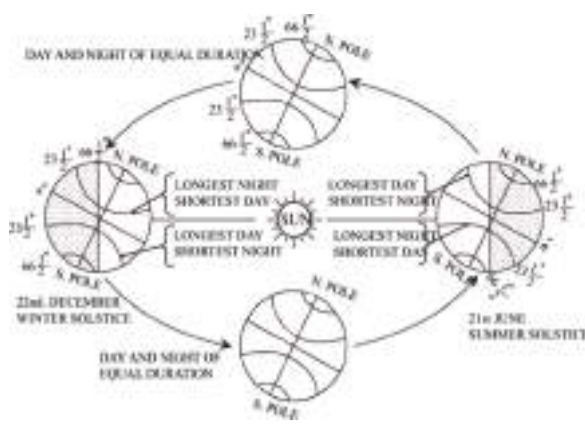


Figure 3.9 : Increase and Decrease of day-night

instead. The day lasts for 24 hours from the Arctic Circle ($66^{\circ}5'$ North) towards the North to the North Pole, and the night lasts for 24 hours from the Antarctic Circle ($66^{\circ}5'$ South) towards the South to the South Pole as well. After June 21 the Sun no longer moves towards north hemisphere, and starts moving towards south hemisphere. This position of the Sun is called Summer Solistice.

September 23: While June 21 becomes over, the North Pole continues to move away from the sun and the South Pole continues to draw near gradually. Consequently, day gradually becomes longer and night shorter in the north hemisphere, and in the south hemisphere, day gradually becomes longer and night shorter. The Earth stays at such a place on September 23, where both the Poles remain equally away from the sun. On this day, the sunlight falls lengthways towards the Equator (at angle 90°), towards both the Arctic and Antarctic circles at angle $66^{\circ}5'$, and towards the Poles at angle 0° . For this reason, day and night on this day remain equal across the Earth.

December 22: While September 23 becomes over, the North Pole continues to move farther away from the sun, rather the South Pole draws nearer. Consequently, the daytime gradually becomes shorter and night longer in the north hemisphere. Once when the Earth stays at such a place on December 22, the South Pole remains most inclined (23.5°) towards the sun. The sunlight falls lengthways (90°) towards the Tropic of Capricorn this day. On this date, the day becomes the longest and the night shortest in the south hemisphere. After December 22 the Sun no longer moves towards south hemisphere, and starts moving towards north hemisphere. This condition of the Sun is called Winter Solistice.

March 21: Subsequently after September 23, when the Earth gets farther ahead to its own orbit, the North Pole gradually draws near the sun, whereas the South Pole moves away instead. For this, day continues to be longer and night shorter in the north hemisphere. At last on March 21, the Earth reaches at such a place on its own orbit, where both the Poles remain equally away from the sun. Like September 23, day and night become equal on this day. This stay of Earth is called the Vernal Equinox. After 21 March, the Earth restores to the previous position of 21 June through moving on its own orbit. The waning and waxing of days and nights happen for the annual motion.

The Change of Seasons

Let us have a look to the adjoining figure. Here while moving round the sun, an explanation regarding the change of seasons has been provided from the four stays of the Earth. The sunrays fall somewhere lengthwise and somewhere else

obliquely because of the annual motion which also results in waning and waxing day and nights. Since the sunrays fall lengthways reach by cleaving less layers of air, they make the surface more heated, and the rays fall Obliquely do not reach by cleaving more layers of air only, but they get extensively spread out in comparison to the rays fall lengthways. Consequently, temperature on all over the surface varies at different times of the year and it results in the change of seasons in the course of time. The difference or transformation of temperature in the Earth in the discrimination of time is called the change of seasons. While moving round the sun, an explanation regarding the change of seasons has been found from the four conditions of the Earth.

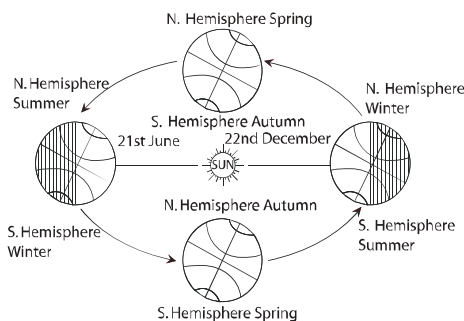


Figure 3.10 : The Change of seasons

The Summer in the North Hemisphere and The Winter in the South Hemisphere:

The 21st day of June is the last day for the summer solistice of the sun. The sunrays fall lengthwise towards the Tropic of Cancer this day. Accordingly here the day becomes the longest and the night shortest. Heat prevails much within the territory of the North Pole for a total period of three months from one and a half months earlier as well as one and a half months later since June 21. During this period, the summer season occurs in the north hemisphere. Since the sunrays fall obliquely during this period for the days being longer and the nights shorter in the south hemisphere, the winter season comes over there.

The Autumn in the North Hemisphere and the Spring in the South Hemisphere: On the 23rd day of September, the sunrays fall lengthways over the Equator and day and night remain equal all over the Earth. For this reason, during a total period of 3 months pervading over one and a half months earlier as well as one and a half months later since the date, heat remains in moderate. During this period, the Autumn occurs in the north hemisphere whereas the Spring in the south hemisphere.

The Winter in the North Hemisphere and The Summer in the South Hemisphere:

The 22nd day of December is the last day for the winter solistice of the sun i.e. the sunrays fall lengthwise towards the Tropic of Capricorn this day. Accordingly, here the day becomes longer and the night shorter. One and a

Individual work: Fill in the table		
Date	N. Hemisphere	S. Hemisphere
June 24		
Sept 25		
Dec. 11		

half months earlier and later since this date, the summer season occurs in the South hemisphere whereas the winter in the north hemisphere for opposite reason.

The Spring in the North Hemisphere and The Autumn in the South Hemisphere:

Both the North and the South Poles stay equally away from the sun on 21 March. The sunrays fall towards the Equator lengthwise in this day, and the day and night become equal everywhere. One and a half months earlier and later since 21 March, the spring season occurs in the north hemisphere whereas the Autumn in the south hemisphere.

The Reasons for the Change of Seasons

As the consequence of the annual motion of the Earth, the sunrays fall somewhere lengthwise and elsewhere obliquely, which result in regular change in temperature and the change of seasons. The annual motion of the Earth causes waning and waxing of the days and nights as well. When the daytime at a place becomes longer than that of night, the atmosphere of that place becomes hotter. In this way, the heat becomes different in the different times of the year all over the earth, and the change of seasons occurs, and the waning and waxing of

Individual work: Identify the reasons for changes of seasons.

day and night occur in the course of time. While moving round the sun, the Earth always rotates inclining at the angle of 66.5° . Consequently, an angular disparity occurs in respect of dropping sunrays to different sites and seasons get changed. The length of the Earth's orbit possesses a measure of 938051827 km. Since this orbit of Earth is elliptical in shape, while rotating, distance of Earth from the sun does not always remain equal. The sun stays closest to the Earth during January 1-3 which is called the Earth's perihelion. In other words, the closest point to the sun in a planet's orbit is called Perihelion. Similarly, the sun stays remotest from the Earth during July 1-4. It is called the Earth's Aphelion. In modern times, Earth's perihelion occurs around January 3, and the aphelion around July 4. It is proved from the decrease and increase of the distance in between the sun and the earth, and from the actual change of the relative volume, it is proved that the Earth's orbit is elliptical in shape. Consequently, the sunbeams vary and seasons get varied as well. Our Bangladesh is situated within the north hemisphere and the Tropic of Capricorn line passes (23.5° North Axis) through the middle of the country. So, the seasons of Bangladesh get changed in accordance with the change in the relative distance of the Sun in the Northern Hemisphere.

Section 3.4: The Ebb and Flow

Apart from the oceanic current in different seas and oceans, the mass-water has a motion of itself. Consequently, the seawater gets inflated for a while every day (the water heights get up), rather it comes to a stop for some while. Such inflation

of the seawater in a regular manner is called the high tide or flow, and disembarking of the same is called low tide or ebb. The ebb and flow mainly occurs due to the earth's motion itself and under the influence of the moon and the sun thereupon. The ebb and flow has diverse categories. Besides, it affects the Earth in various angles and aspects.

The Surmise of the Ebb and Flow, their Reasons and Classifications

The sun and the moon continuously attract the surface of the Earth. The water of its surface gets inflated because of such attraction at a site and gets descended elsewhere at intervals of a particular duration, sea-water regularly ascends and descends once after every 12 and a half hours in due course. Both the ebb and flow

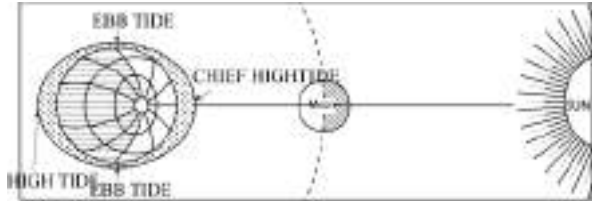


Figure 3.11 : The Ebb and Flow

successively occurs after every 6 hours and 13 minutes. Such inflation of the seawater in a regular course is called the high tide or flow, and descending there of is called the low tide of ebb. The water at the middle-sea generally ascends high or descends below by 1-3 feet only. But because of the depth of the seas and oceans at littoral regions, the mass-water ascend and descend very high and low. That is why the ebb and flow is observed few kilometers from the estuary throughout the rivers' course.

The Reasons of the Ebb and Flow: In the ancient period, different fanciful suppositions about the reasons of tide were customary, but nowadays the scientists have proved that the high and low tides occur in consequence of the centric force generated due to the Earth's rotation and of attraction from the sun and the moon over the Earth.

The Influence of Gravitation Force: Each substance of the Earth has an attraction, and one attracts the other. This attraction is called the gravitation force or the gravity. Due to the gravity, the Earth moves round the sun, and the moon moves round the Earth. One that is so large possesses the gravitation force so much relatively. But when the distance becomes far, the gravity becomes weak. Though the sun is larger in size than the moon by 2.60 crore-fold, but the distance of the sun from the Earth is comparatively much more than the distance of the moon, the attraction force of the moon over the Earth is more than that of the sun by twofold. Thus the moon's attraction causes generating the ebb and flow.

The Centrifugal Force: Since the Earth staying on its own axis or spine moves around rapidly, liquid mass-water has the tendency of getting scattered from its surface. Such force is named Centrifugal Force. Due to the rotation of the earth and the moon, the influence of the Centrifugal Force over the liquid and light

mass-water of the surface prevails extensively. For this reason, the mass-water gets ejected outside all the while and trends to become isolated from solid ground of the liquid mass-water. In the same manner, the Centrifugal Force helps originate the ebb and flow.

The Classifications of the Ebb and Flow: The ebb and flow or tide may be divided into a few categories such as the Major Tide, the Minor Tide, the Flood Tide or the Spring Tide and the Neap Tide or the Ebb Tide. The moon moves round the Earth. During the rotation of the moon, the part of the Earth that draws near the moon, the moon's attraction prevails mostly over there. Consequently, water arrives from the neighboring sites and gets inflated just below the moon, and tides originate. This tide is categorically called Major Tide, or in other words, direct tide. To the opposite of the Major Tide, the landsite below the water is firmly tied with the Earth's centre, because of which an attraction of the moon over there remains equal in comparison with that of the centre. Consequently, the landsite becomes more attracted towards the moon than the water-site of the opposite side. Resultantly, the Centrifugal Force is generated. Water from both the sides flows towards that extent, and causes the generation of tide. Such tide is called Minor Tide or Indirect Tide. When a major tide originates at one side of the Earth, and a minor tide at the other side, water flows out of the middle between the two tides. The state of water existing at the middle is called the ebb.

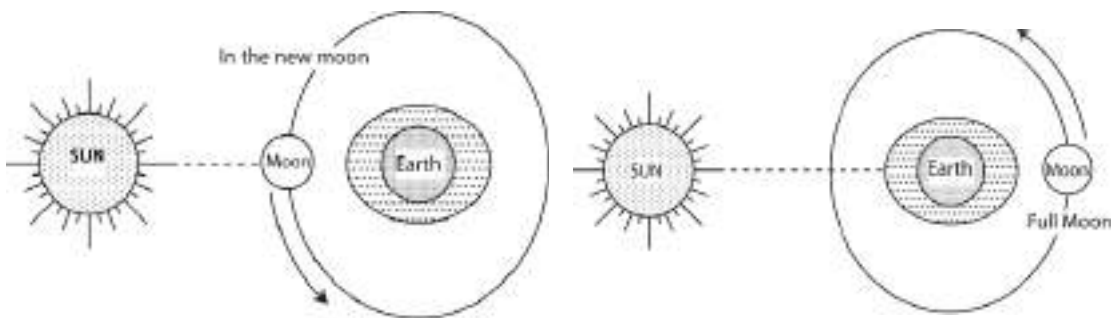


Figure 3.12 : Flood tide (in the duration of new moon) Figure 3.13 : Flood tide (in the duration of full moon)

On a new moon lunar day, the moon and the sun stay at the same side of the Earth, and on a full moon day, the moon stays at one side of the Earth whereas the sun stays at the other side instead. Consequently, the sun and the moon stay at the prime vertical circle during these two lunar days, and the tide that strongly get generated due to their combined attraction, is called the Flood Tide or the Spring Tide.

Since the sun and the moon stay at a right angle of the Earth on the 7th and 8th lunar days, the tides originate towards the moon because of its attraction. But these tides' velocity does not get the severity because of the sun's attraction at the same time. Such tide is called the Neap Tide or the Ebb Tide. The Prime Tide originates twice and the Neap Tide also twice a month.

The Interval of the Ebb and Flow

As the Earth rotates upon its own axis from the West to the East so the moon orbits round the Earth from the West to the East too. The moon moves round the Earth on its own orbit once in 27 days, so the moon passes its regular route ($360/27$) or 13° in the duration of its one complete rotation around. As both the sun and the moon rotate from the West to the East, the Earth

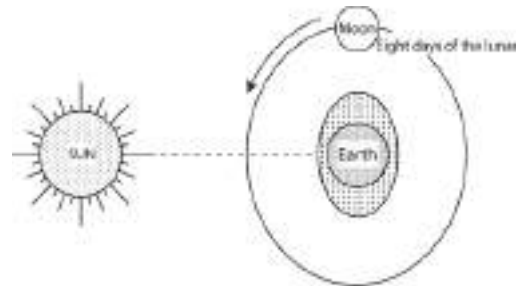


Figure 3.14 : Ebb-tide on the 8th day

passes through the said 13° in further $(13 \times 4) = 52$ minutes. So, when the Major Tide originates at any certain site at a particular duration, a minor tide also originates over there after 12 hours and 56 minutes from the originating time of the major tide, and again a major tide originates over there after 24 hours and 52 minutes from the time of the major one. So, at every concerned site, there originates an ebb after 6 hours and 13 minutes from the originating time of a flow in due course.

The Influence of the Tides on the Earth

The tide has an influence directly or indirectly on the Earth or the ground-surface, the mass-water and the economic activities of human beings. Due to occurrence of the ebb and flow twice a day, the sweepings existing in the river get cleansed and its water becomes

Individual work: Describe the reasons for the ebb and flow and their influence on the Earth.

free of filths or dirt, and silts gathering at the river-channel and river-mouths hinder the flow. The river-pits become deeper due to the current of ebb and flow. The tide of water is kept confined by cutting canals beside a number of rivers for the irrigation purpose. Hydro-electricity is generated at lots of rivers in the world by utilizing the tide such as the Larance Power Plant in France, and a hydro-electricity center at Bandala Port in India. As salty water inundates the interior side of the country by the ebb and flow, the river water movement remains favorable in the countries characterized by the predominance of cold. Due to the increase of river water, big ships, vessels to the ocean can easily enter the rivers;

again, they can go back to the ocean with the ebb. In Bangladesh while the depth of river extends at the Chittagong and the Mongla ports during tide, big vessels can easily enter or leave the port. The vessels, ships, etc. cast their anchors at the river-channel before entry to the port waiting for tide. The tide from the Bay of Bengal reaches near Goalanda at the river Padma and at the Meghna near Bhairab Bazar. Salt is also produced from the sea-water keeping confined through drying process. The sea-water largely enters the land-site passing through the river-channel during the spring tide in strong waves, and causes Tidal bore. Tidal bore holds the height of 3-4 feet to approximate 40 feet. The rivers whose estuaries are narrow or facing sand damp in their front, originate strong tidal bore. In Bangladesh, strong tidal bore usually takes place in the new moon's tide during the rainy season. But its velocity gets increased after its entrance into the land surface. The Meghna, the Vagirothi, the Amazan, etc. rivers become usually succumbed with the excessive tidal bore. A havoc wear and tear including submersion of boats, steamers, ships, etc. inadvertently happen due to such tidal bore from time to time.

Activity: The teacher will create a checklist on various topics in this chapter, such as local time, tides, seasonal changes, etc. Students will answer. The teacher will provide necessary feedback.

Exercise

Multiple Choice Questions

1. Which one is a satellite of Uranus?
 - a. Capitatus
 - b. Ariel
 - c. Nereid
 - d. Ganymede
2. Due to the distance of planets from the Sun
 - i. the time of revolution will differ
 - ii. The temperature will differ
 - iii. The number of satellites will vary.

Which of the following is correct?

- a. i and ii
 - b. i and iii
 - c. ii and iii
 - d. i, ii and iii
3. Due to the Earth revolving in an elliptical orbit-
 - i. Different seasons appear.
 - ii. Gravitational force fluctuates.
 - iii. The length of day and night varies.

Which of the following is correct?

- a. i and ii b. i and iii
c. ii and iii d. i, ii and iii

Read the following passage and answer questions 4 and 5

Arpita sits down to study very early every morning. One day, she notices a star visible in the eastern sky even in the early morning. Arpita understands that she has seen a planet.

4. What is the name of the planet seen by Arpita?
- a. Venus b. Saturn
c. Mars d. Neptune
5. What dissimilarity will be observed between the Earth and the said planet?
- a. The planet has no satellites.
b. The planet's temperature is below freezing point.
c. The planet has rings around it.
d. The planet is bluish in color.

Creative Questions

1.

Place	Latitude	Longitude	Date	Time
A	30° North	105° West	June 22 nd	7 am
B	50° South	56° West	June 22 nd	?

- a. What is an axis (or polar axis)?
b. What is a sunspot? Explain.
c. If the local time at location A in the table is 7 AM, what will be the local time at location B?
d. Will the length of day and night be the same in both locations on that date? Present your arguments in support of your answer.

2. Maisha lives in Sweden (66.50° North latitude and 15° East longitude). On September 25th, she wishes her younger sister Maliha, who lives in Canberra (35° South latitude and 150° East longitude), a happy birthday at 6 AM local time in Sweden. Maliha mentions in passing that she will visit Sweden on December 28th.

- What is a solar day?
- Why do airplanes fly through the tropopause?
- What time in Canberra did Maisha make the call?
- Will the two locations experience the same season on Maliha's travel date?
Provide your arguments based on the given information.

3. Sinthia goes to Cox's Bazar with her parents. In the evening, they are captivated by the calm sea under the full moon. A few moments later, they notice the sea level rising and crashing violently against the shore. Her father reassures her not to be scared and says that such conditions regularly occur in the sea.

- How many types of tides are there?
- What is a leap year? Explain.
- Explain the reason for the reaction in the seawater at that time.
- Does the event Sinthia witnessed have an impact on human economic activities? Analyze.

4.

Place	Latitude	Longitude	Date	Time
Mexico	30° North	105° West	June 22 nd	?
Willing Island	50° South	75° West	June 22 nd	7 AM

- What is an antipode?
- What is perihelion? Explain.
- Determine the time difference between the local time of Willing Island and Mexico as mentioned in the table.
- Analyze the comparative length of day and night in the two locations on that date.

Short Answer Questions

- What is the importance of the International Date Line?
- Why is the Earth suitable for the habitation of animals and humans? Explain.
- What is the reason for the high temperature of the ozone layer?